

Heart Disease Prediction by using data Mining

a) Problem Statement:

One of the biggest leading causes of death within worldwide healthcare system is due to heart disease. Therefore, early recognition is crucial since it helps timely interventions, and it can also prevent brutal outcomes. Nowadays, in medical science, there is high availability of patients' clinical data, therefore machine learning, and data-mining techniques gives an opportunity in identifying patterns of heart disease.

This project explores whether heart disease can be predicted accurately by using patient's clinical indicators through machine learning models. The aim is to test the predictive model and explore how such tools could support healthcare decision making and early screening of the disease.

b) Data Summary:

The dataset used in this project is “Cleveland subset of the UCI Heart disease dataset”, acquired from the UCI Machine Learning Repository. The dataset is consisting of 303 patients' clinical data, with 14 attributes including cholesterol levels, age, sex, Thallium stress test result, maximum heart rate, resting blood pressure, and other medical related indicators.

The target column in the attributes indicates the presence of heart disease. In the original dataset form target variable shows the multiple levels of disease severity but for this analysis target variable was converted into binary form representing the presence (1) and absence (0) of heart disease. This transformation allows us to frame the problem as binary classification task, which is fully suitable for predictive healthcare modelling.

At start the dataset appeared as fully complete with zero null values but after some analysis it revealed that the missing values were encoded using “?” symbol in some variable columns. For accurate model training these values required preprocessing before analysis.

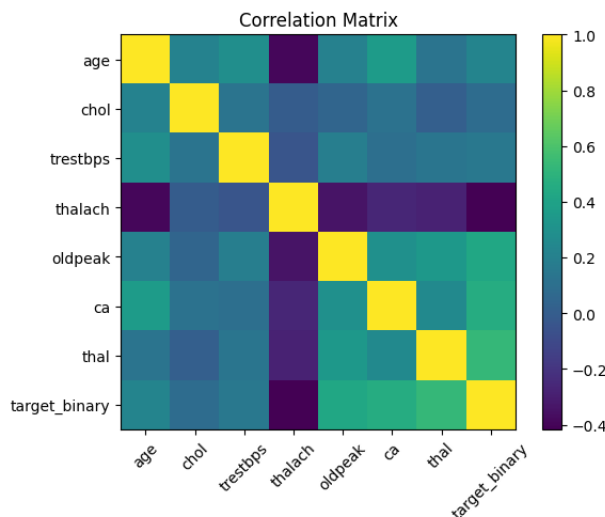


Fig 1: Correlation matrix, highlighting relation between heart diseases presence and clinical indicators such as oldpeak, ca and thal suggesting these features are informative for prediction

c) Methods:

In this study, supervised data mining techniques are used to predict heart disease. Before modelling, the dataset go through preprocessing, including conversion of missing data “?” into null values and to preserve all patients' data we use median imputation for filling null values. To ensure that variables contributed equally for model performance feature scaling is also used.

Two classifications are used for training and evaluation:

1. Logistic regression: this model is used as a primary model for its interpretability and extensive use in healthcare research.
2. Decision tree classifier: this model is used as comparison model to inspect non-linear relationship within the data

At first the dataset was split into testing and training subsets using stratified sampling. Accuracy, recall, F1 score and precision was used to evaluate model performance. a confusion matrix is used to check the classification performance emphasis on recall which is very important in identifying at risk patients. Furthermore, a baseline model is also used which always predict the most common class to access the machine learning model is giving meaningful insights.

d) Results:

The logistic regression model shows a high predictive performance, by providing an accuracy of 86.8% and achieved a precision of 81.2%. It shows that most patients who have been predicted as experiencing heart disease were correctly identified. Above all the model achieved a recall of 92.8% which means that the model correctly identified most patients who truly had a heart disease. The F1 score achieved by the model is 86.6% which indicate a fit balance between precision and recall.

The second model trained on the data is decision tree, achieved an accuracy of 78.6% which is lower and less effective than the logistic regression model. Decision tree model can capture complex patterns, so the logistic regression model demonstrated as more suitable for this case.

The baseline model achieved an accuracy of 54.1% which always predict the most common class. The improvements achieved by logistic model confirms that machine learning models can learn meaningful patterns from patients' clinical data.

Results proves that logistic regression is a reliable and effective method for heart disease risk prediction using patient's data

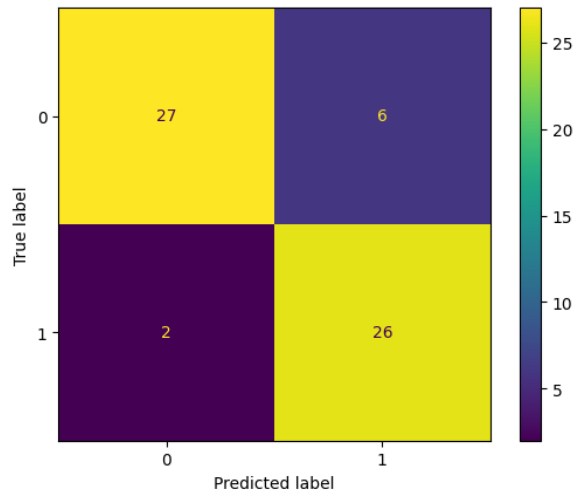


Fig 2: logistic regression model confusion matrix, illustrates model's ability to correctly identifying patients with heart disease.

e) **Conclusions/Recommendations:**

This study illustrates that by using machine learning methods, especially logistic regression model, can predict heart disease presence using patient's clinical data. Logistic regression model achieved a high recall which is very important in healthcare context, because failure to spot at risk patients can delay diagnosis and treatment.

Based on this study findings, it is recommended that healthcare system should use interpretable machine learning models such as logistic regression models for early disease screening processes. To prioritise further diagnostic testing these models can be used as decision support tools to help doctors identifying high risk patients.

However, this dataset originates from a single population and relatively small so the model may have some limitations. For future work recommendations are testing the model on more diverse and larger datasets, also exploring more algorithm's and doing further model tuning to get better performance and results from the machine learning model.

In conclusion this study shows the potential of data mining techniques for early detection of heart disease and better healthcare outcomes when used with clinical expertise.

References:

Janosi, A., Steinbrunn, W., Pfisterer, M. and Detrano, R. (1988). *UCI Machine Learning Repository*. [online] archive.ics.uci.edu. Available at: <https://archive.ics.uci.edu/dataset/45/heart+disease>.